

Olist E-Commerce Sales Analysis Report

Objectives

The primary objective of this project is to analyse the Olist Brazilian E-Commerce dataset and transform raw transactional data into meaningful business insights through data preprocessing, exploratory data analysis, feature engineering, and interactive dashboard development.

Data Set

olist_customers_dataset.csv
olist_geolocation_dataset.csv
olist_order_items_dataset.csv
olist_order_payments_dataset.csv
olist_order_reviews_dataset.csv
olist_orders_dataset.csv
olist_products_dataset.csv
olist_sellers_dataset.csv
product_category_name_translation.csv

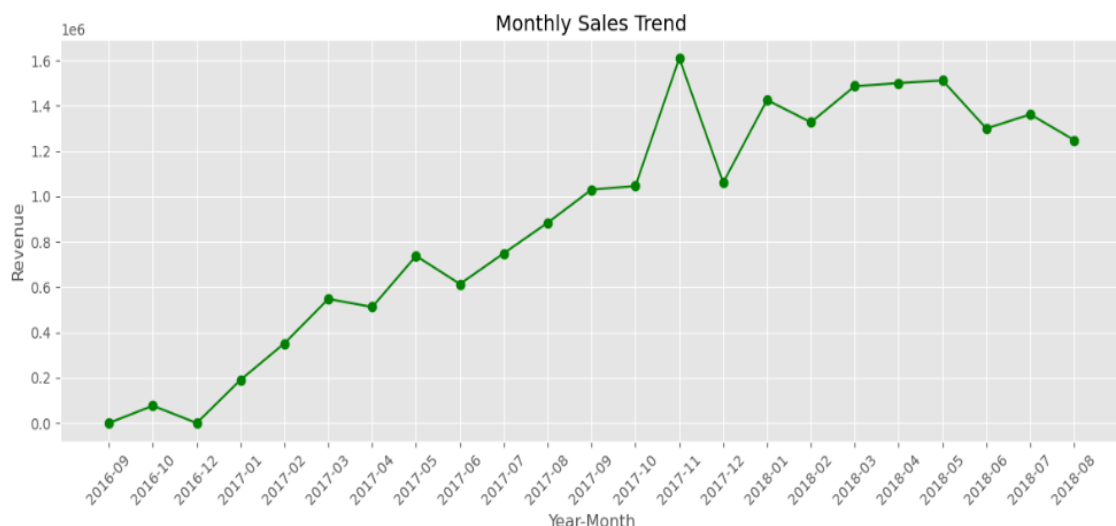
Data Cleaning

- Verified primary and composite keys across all tables.
- Converted date fields to appropriate datetime format.
- Investigated missing values and distinguished valid business scenarios from genuine data quality issues.
- Confirmed the absence of duplicate records in most tables; duplicate review IDs were validated as legitimate business records.
- Flagged minor business rule violations (23 records, 0.02%) without altering the original data.
- Retained products with missing category information to preserve revenue while excluding them from category-level analyses.

Outcome: The cleaning process ensured high data quality while preserving business integrity for subsequent analysis.

Exploratory Data Analysis (EDA)

1. Analyse monthly sales trends.



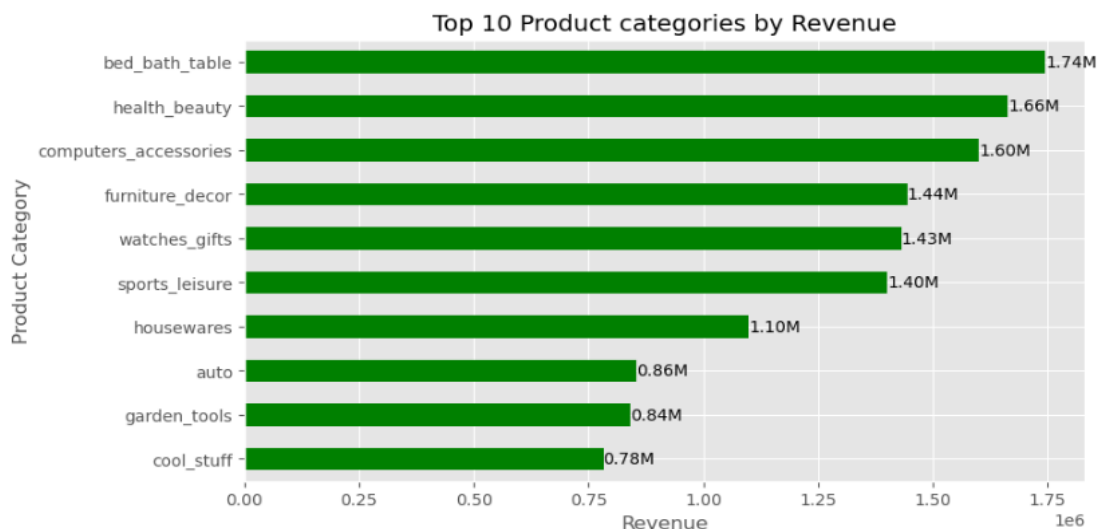
Observation

- Revenue shows a strong upward trend over the analysis period, reaching its highest level in November 2017 (~1.6M). Although monthly sales fluctuate, revenue remains consistently high throughout 2018.

Business Insight

- The sustained growth indicates increasing customer demand and business expansion. The sales peak in November suggests seasonal demand, highlighting the opportunity to strengthen inventory planning and marketing campaigns during high-sales periods.

2. Product Category Analysis



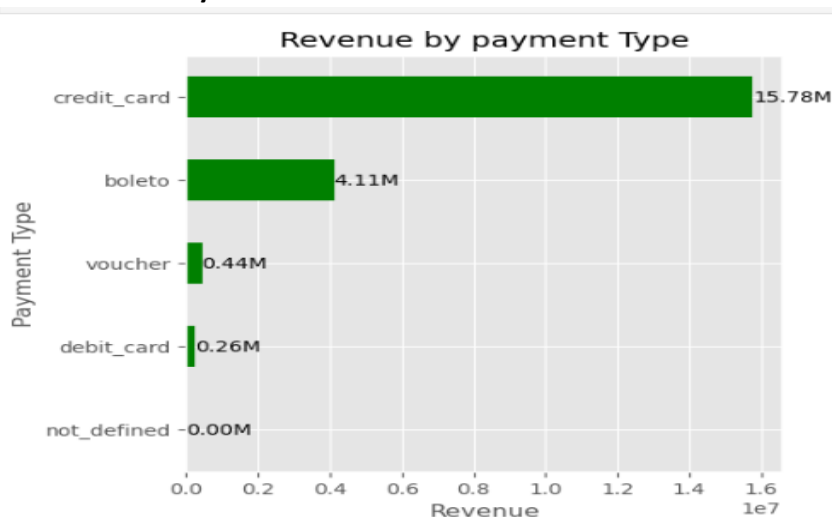
Observation

- Bed, Bath & Table generated the highest revenue (1.74M), followed by Health & Beauty (1.66M) and Computers & Accessories (1.60M). These categories were the top revenue contributors among the top 10 product categories.

Business Insight

- The business relies heavily on a few high-performing product categories for revenue. Prioritizing inventory, marketing, and promotional efforts for these categories can help maximize sales while identifying opportunities to improve lower-performing categories.

3.Payment Method Analysis



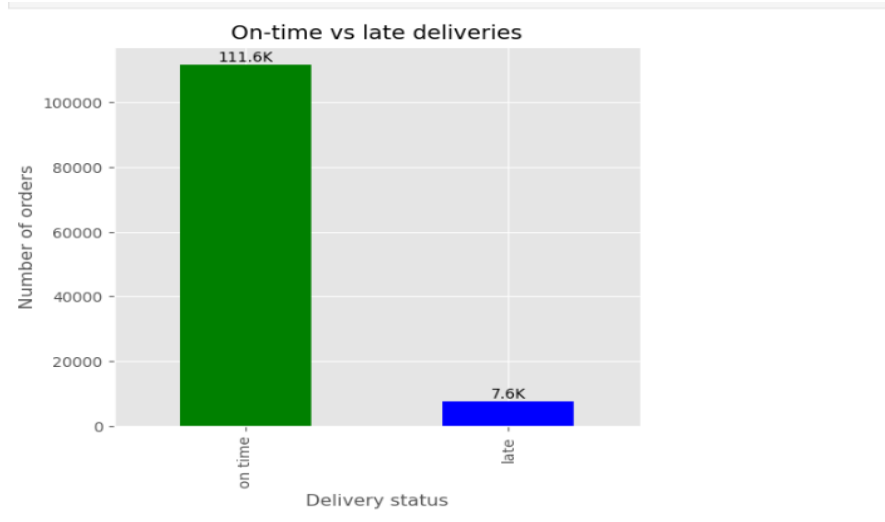
Observation

Credit cards generated the highest revenue (15.78M), significantly outperforming all other payment methods. Boleto ranked second (4.11M), while voucher and debit card contributed only a small share.

Business Insight

Customers show a strong preference for credit card payments, making it the primary revenue driver. Optimizing the credit card payment experience and offering card-based promotions could help maximize sales.

4.Delivery Performance Analysis



Observation

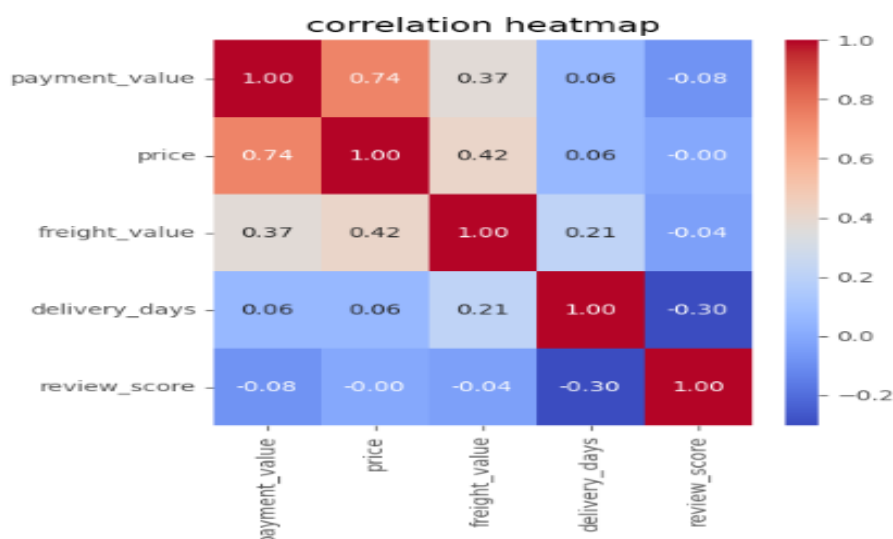
- Approximately 111.6K (93.6%) orders were delivered on time, while only 7.6K (6.4%) were delivered late, indicating that the majority of deliveries met the expected delivery timeline.

Business Insight

- The high on-time delivery rate reflects an efficient logistics process and contributes positively to customer satisfaction. Further analysis of the small proportion of late deliveries can help identify bottlenecks and improve delivery performance.

5.Correlation Analysis

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59]: <function matplotlib.pyplot.show(close=None, block=None)>
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Observation

- **Payment value** has a strong positive correlation with **price (0.74)** and a moderate positive correlation with **freight value (0.37)**. **Delivery days** show a moderate negative correlation with **review score (-0.30)**, while the remaining relationships are weak.

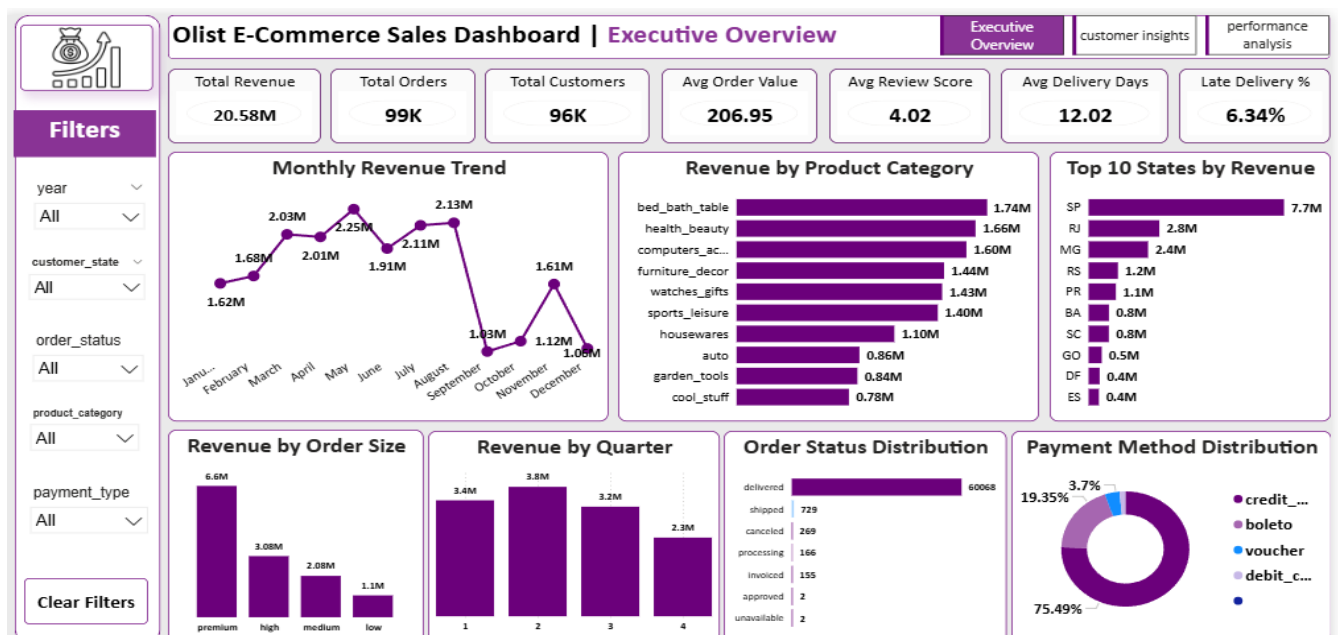
Business Insight

- Order value is primarily influenced by product price, whereas longer delivery times are associated with lower customer satisfaction. Reducing delivery delays could help improve customer review scores and enhance the overall customer experience.

Feature Engineering

Feature Category	Features Created	Business Purpose
Delivery Performance	delivery_days, shipping_days, carrier_to_customer_days, delivery_delay, is_late_delivery	Measure delivery efficiency and identify delayed orders.
Order Processing	approval_hours	Evaluate the time taken to approve customer orders.
Time Intelligence	purchase_year, purchase_month, purchase_month_name, purchase_day, purchase_quarter, is_weekend	Analyze seasonal, monthly, quarterly, and weekday purchasing patterns.
Customer Reviews	review_category	Categorize customer feedback into Negative, Neutral, and Positive sentiments.
Financial Metrics	total_item_value, freight_perc	Analyze total order value and the contribution of freight charges to product cost.
Order Segmentation	high_value_order, order_value_category	Identify high-value orders and classify orders into Low, Medium, High, and Premium value segments.

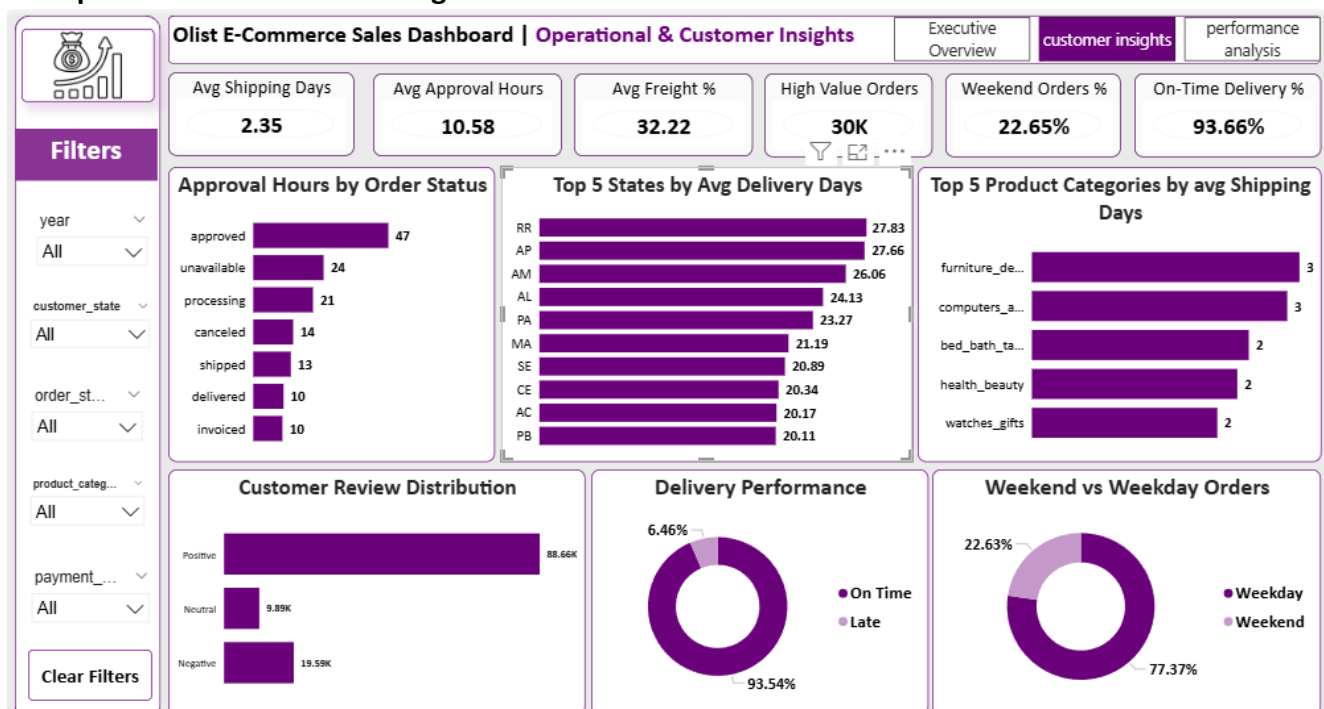
Power BI Dashboard – Executive Overview



Key Business Insights

- The business generated a total revenue of 20.58M from approximately 99K orders, with an average order value of 206.95.
- Bed, Bath & Table, Health & Beauty, and Computers & Accessories were the top three revenue-generating product categories.
- São Paulo (SP) contributed the highest revenue (7.7M), followed by Rio de Janeiro (2.8M) and Minas Gerais (2.4M).
- Revenue showed an overall upward trend, with peak sales observed during May–August, indicating periods of high customer demand.
- Premium-value orders generated the highest share of revenue compared to other order segments.
- Quarter 2 (Q2) recorded the highest revenue among all quarters, reflecting strong business performance during this period.
- Credit card was the most preferred payment method, accounting for approximately 75% of all transactions.
- The business maintained strong customer satisfaction with an average review score of 4.02, while achieving an average delivery time of 12.02 days and a low late delivery rate of 6.34%.

Operational & Customer Insights

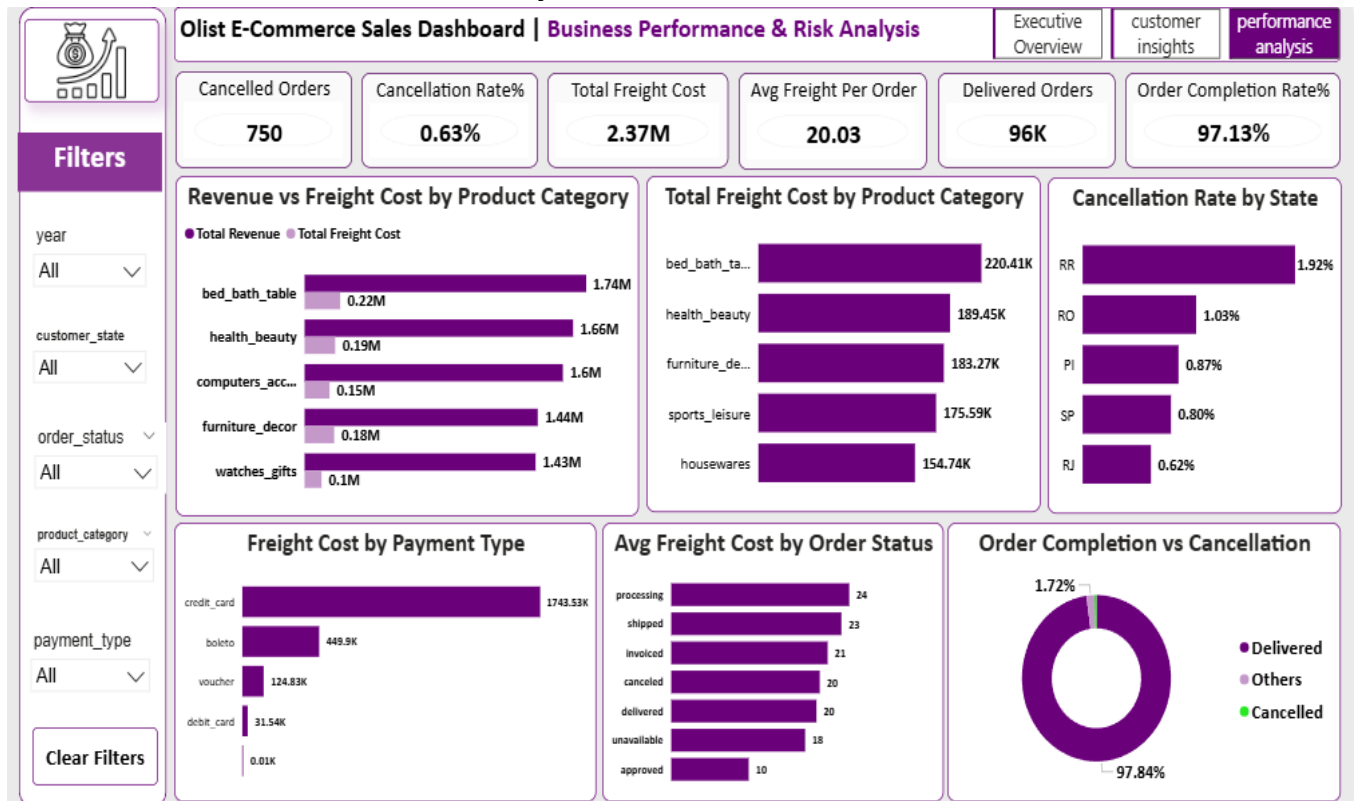


Key Business Insights

- The business maintained strong operational efficiency with an **average shipping time of 2.35 days**, **average order approval time of 10.58 hours**, and an **on-time delivery rate of 93.66%**.
- Approximately **30K orders** were classified as **high-value orders**, indicating a significant contribution from premium purchases.
- **Positive customer reviews (88.66K)** significantly outnumbered neutral and negative reviews, reflecting high customer satisfaction.
- **Weekday orders (77.37%)** were considerably higher than **weekend orders (22.63%)**, suggesting that most purchases occurred during the workweek.

- States such as **Roraima (RR)**, **Amapá (AP)**, and **Amazonas (AM)** recorded the highest average delivery times, highlighting potential opportunities for logistics optimization.
- Product categories including **Furniture & Decor** and **Computers & Accessories** required relatively longer shipping times, indicating that product characteristics may influence fulfilment duration.

Business Performance & Risk Analysis



Key Business Insights

- The business achieved a 97.13% order completion rate with only 750 cancelled orders (0.63%), indicating strong operational performance.
- Bed, Bath & Table generated the highest revenue (1.74M) while also incurring the highest freight cost (220.41K), followed by Health & Beauty and Computers & Accessories.
- The total freight cost amounted to 2.37M, with an average freight cost per order of 20.03, highlighting logistics as a significant operational expense.
- Credit card payments accounted for the largest share of freight costs, reflecting their dominance as the preferred payment method.
- Roraima (RR) recorded the highest cancellation rate (1.92%), indicating potential regional challenges that may require further investigation.
- The order completion vs. cancellation analysis shows that 97.84% of orders were successfully delivered, demonstrating an efficient order fulfilment process with minimal cancellations.

Key Findings

- The business generated 20.58M in revenue from 99K orders.
- Bed, Bath & Table was the highest revenue-generating product category.
- Credit cards accounted for the majority of transactions and revenue.
- Sao Paulo contributed the highest revenue among all states.

- Approximately 93.66% of orders were delivered on time.
- The average customer review score was 4.02, indicating high customer satisfaction.
- Quarter 2 recorded the highest revenue.
- Only 0.63% of orders were cancelled, resulting in a 97.13% order completion rate.

Business Recommendations

- Increase inventory for top-performing product categories to meet demand.
- Optimize logistics in states with longer delivery times.
- Monitor regions with higher cancellation rates and investigate underlying causes.
- Continue promoting credit card payment options while encouraging adoption of alternative payment methods.
- Use seasonal sales trends to improve inventory planning and marketing campaigns.
- Continuously monitor delivery performance to maintain high customer satisfaction.

Conclusion

This project demonstrated an end-to-end data analytics workflow using Python and Power BI. Through data cleaning, exploratory data analysis, feature engineering, and dashboard development, meaningful insights were generated regarding sales performance, customer behaviour, logistics, and business operations.